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To the Editor-in-Chief and Associate Editor
IEEE Transactions on Information Theory

Dear Editor:

Please consider the enclosed manuscript, “Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context,” for publication in the IEEE Transactions on Information Theory. I am the sole author.

The paper evaluates Steinberg’s common-reconstruction functional on an augmented Gaussian source: the encoder observes a pair (Y, V) , the distortion binds Y alone, and a third party holds a noisy copy of the context V that no decoder uses. Three results are primary: a closed form for the minimal conditional content, the larger root of an explicit quadratic, for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source; the exact region of jointly achievable rate and conditional content, with a two-water-level Pareto frontier and a strict tradeoff exactly when the described variable and the context are partially correlated, persisting at the clean-context boundary; and non-determination: the function is not a functional of the reduced source-side-information marginal, so the encoder’s access to the context changes the operational problem. Consequences and extensions settle the exact attainment of the determinant lower bound, the frontier with vector context, the complete binary solution through a tilt equation with its full rate–content frontier, and a direct Gaussian operational theorem. Consistency anchors recover the classical, Gray, and Steinberg corners.

Two earlier documents are relevant, and they are distinct from one another. First, this Transactions recently declined a different manuscript of mine, a broad synthesis. I ask the editors to read the present submission independently of it: this is a single-object work in the classical rate–distortion genre and shares no theorem with that declined submission. The terminology concerns raised in that review have been addressed here by adopting standard source-coding vocabulary throughout.

Second, and separately, the manuscript cites an archived preliminary manuscript by the author (Zenodo record, doi:10.5281/zenodo.21776291), as disclosed in its Related Work section. The discrete operational region theorem and its single-letterization appeared there, as did the scalar Gaussian corner, and both are inherited here with attribution. Everything from Section III onward is new, and the present paper supersedes the source-coding content of that record, which remains the archived evidence artifact.

All closed forms are verified by an apparatus available as supplementary repository material: symbolic and numerical verification scripts, including two written independently of the derivations, and a partial Lean 4 formalization of the load-bearing algebra.

This manuscript is not under review elsewhere, and I have no conflicts of interest to declare.

Sincerely,

Andrew H. Bond
Senior Member, IEEE